Numerical Solution of a Variable-Diffusion Elliptic BVP

Project 2 (PDE BVP) Report — Team 15

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Abstract

Steady heat conduction in a heterogeneous medium is modelled by the divergence-form elliptic problem $-\nabla \cdot (a\nabla u) = f$ on the unit square (Problem Pack Direction 3). I discretise with a conservative face-averaged five-point finite-difference scheme, assemble in COO, solve in CSR, and move Dirichlet data into the right-hand side so that boundary nodes are never unknowns. On the smooth manufactured benchmark $a = 1 + x$, $u^* = \sin(\pi x)\sin(\pi y)$, discrete L^2 and L^∞ errors both fit order 2.00 over $n = 15, 31, 63, 127$, with $\|A - A^\top\|_{\max}/\|A\|_{\max} = 0$. A barrier-function argument proves $\|u_h - u^*\|_\infty \le \|\tau_h\|_\infty/4$ for this problem, and an eigenvalue bound predicts the discrete L^2 error to within 10%. On the layered jump $a = 1\,|\,10$ at $x = 1/2$, a harmonic face average is provably nodally exact ($L^\infty \sim 10^{-14}$) and recovers the interface flux 20/11; for the arithmetic average a one-dimensional flux argument predicts the first-order L^∞ error to within 0.2%, and the maximum flux jump stays $\mathcal{O}(1)$ at the interface–boundary corners. Unpreconditioned CG matches the direct discrete L^2 error once the relative residual reaches 10^{-4}. At matched accuracy over $n = 31$–511, CG is cheapest only for $n \le 63$, `spsolve` from $n = 127$ on, and AMG-preconditioned CG needs 4–8 iterations on every mesh but its setup keeps it level with `spsolve` at $n = 511$. The scheme is second-order for smooth a; across an unresolved jump an arithmetic face coefficient loses both order and flux continuity, and CG iteration counts grow with contrast a_R/a_L.

Keywords: elliptic BVP, variable diffusion, five-point stencil, harmonic average, method of manufactured solutions, sparse direct / CG

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1 Introduction and Motivation

1.1 Problem Background

Steady heat conduction in a layered composite, soil profile, or tissue boundary is the model problem in which the conductivity $a(x, y)$ varies in space and may jump [8, 1]. The physically correct operator is the divergence form $-\nabla \cdot (a\nabla u)$: the normal flux $a\partial_n u$ is continuous across an interface. A pointwise expansion $a\Delta u + \nabla a \cdot \nabla u$ is the wrong starting point once a is discontinuous—it loses conservation, symmetry, and the M-matrix sign pattern.

I chose Direction 3 because the unit square keeps the five-point machinery of Tutorial 5 intact, while the jump forces a modelling choice (arithmetic versus harmonic face coefficients) that a smooth manufactured Poisson problem never sees.

1.2 Mathematical Model

On $\Omega = (0, 1)^2$ with Dirichlet data on $\partial\Omega$,

$$-\nabla \cdot \big(a(x, y)\nabla u\big) = f \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega, \tag{1}$$

with $0 < a_{\min} \le a(x, y) \le a_{\max} < \infty$. The sign convention makes $-\nabla \cdot (a\nabla u)$ a positive operator, so the discrete matrix has a positive diagonal.

Smooth manufactured benchmark (course bank problem P4).

$$a(x, y) = 1 + x, \qquad u^*(x, y) = \sin(\pi x)\sin(\pi y), \qquad g = 0,$$

and

$$f = 2\pi^2(1 + x)\sin(\pi x)\sin(\pi y) - \pi\cos(\pi x)\sin(\pi y). \tag{2}$$

The extra $-\pi\cos(\pi x)\sin(\pi y)$ term is $\nabla a \cdot \nabla u^*$; a constant-coefficient five-point stencil cannot capture it.

Layered-interface benchmark.

$$a(x) = \begin{cases} 1, & x < 1/2, \\ 10, & x > 1/2, \end{cases} \qquad f = 0,$$

and $g = u^*$ on $\partial\Omega$, where the exact one-dimensional profile

$$u^*(x) = \begin{cases} \dfrac{20}{11}\,x, & 0 \le x \le 1/2, \\ \dfrac{2}{11}\,x + \dfrac{9}{11}, & 1/2 \le x \le 1, \end{cases} \tag{3}$$

is independent of y. It is continuous, $u^*(0) = 0$, $u^*(1) = 1$, the derivative jumps, and the flux $au^*_x = 20/11$ is continuous.

Expected signs: on the layered problem, $u^* \in [0, 1]$ with extrema on the boundary (maximum principle). On P4, $u^* > 0$ in Ω and $u^* = 0$ on $\partial\Omega$.

1.3 Overview of the Report

Section 2 derives the face-averaged stencil and Dirichlet shift, states truncation assumptions, proves max-norm and L^2 convergence bounds for P4, and proves exactness of the harmonic and first-order behaviour of the arithmetic interface rule. Section 3 records the implementation. Section 4 verifies P4

by MMS, separates algebraic from discretisation error, falsifies "symmetry ⇒ correct interface," and reports cost at a matched accuracy target. Section 5 summarises the findings.

2 Discretisation and Analysis

2.1 Discrete Scheme

I use a uniform nodal grid with n interior nodes per direction,

$$h = \frac{1}{n+1}, \qquad x_i = (i+1)h, \qquad y_j = (j+1)h, \qquad i, j = 0, \ldots, n-1. \tag{4}$$

The unknown vector is indexed by $k = in + j$ (`meshgrid` with `indexing='ij'`; axis 0 is x). Boundary nodes are not unknowns.

A conservative finite-volume reading of $-\nabla \cdot (a\nabla u)$ on the dual cell of width h gives the five-point divergence-form stencil [2, 8]

$$(A_h U)_{ij} = \frac{1}{h^2}\Big[a_E(U_{ij} - U_{i+1,j}) + a_W(U_{ij} - U_{i-1,j}) + a_N(U_{ij} - U_{i,j+1}) + a_S(U_{ij} - U_{i,j-1})\Big]. \tag{5}$$

When $a \equiv 1$ this recovers the classical five-point Laplacian

$$(-\Delta_h u)_{ij} = \frac{4u_{ij} - u_W - u_E - u_S - u_N}{h^2}.$$

The four face coefficients are shared: the east coefficient of node (i, j) is the west coefficient of node $(i+1, j)$. That sharing makes A_h symmetric after Dirichlet elimination.

For smooth a I evaluate the coefficient at the half-point, $a_E = a(x_i + h/2, y_j)$. For $a = 1 + x$ this coincides with the arithmetic nodal midpoint and is second-order. At a material interface I use the resistance (harmonic) average of the two nodal values,

$$a_E = \frac{2a_L a_R}{a_L + a_R}, \tag{6}$$

and, as the depth comparison, the arithmetic mean $(a_L + a_R)/2$. When the interface bisects a grid edge, (6) is exact for piecewise-constant a. I therefore take odd interval counts $N = n + 1 \in \{31, 63, 127, 255\}$ on the layered test, so $x = 1/2$ lies midway between two interior columns [8].

2.2 Consistency and Truncation Error

For a single east face with smooth a and u, the centred flux

$$a_E \frac{U_{i+1,j} - U_{ij}}{h} = a\big(x_i + \tfrac{h}{2}, y_j\big)\; u_x\big(x_i + \tfrac{h}{2}, y_j\big) + \mathcal{O}(h^2)$$

is second-order; summing four such faces and dividing by h recovers $-\nabla \cdot (a\nabla u) + \mathcal{O}(h^2)$. The same u_{xxxx} and u_{yyyy} pieces appear as in the five-point Laplacian, plus smooth lower-order terms from ∇a. Assumptions: $u \in C^4(\overline{\Omega})$ and $a \in C^2(\overline{\Omega})$. Across the layered jump, u^* is only Lipschitz and $a \notin C^2$, so this expansion fails at the interface face; second-order truncation is not claimed for an arithmetic coefficient there.

I measure the truncation residual $\tau_h = b - A_h u^*$ in the discrete L^2 norm $h\|\tau_h\|_2$. On P4 this quantity drops by four under each mesh doubling (Table 3), consistent with $\mathcal{O}(h^2)$ pointwise truncation. On the four meshes of Table 2 there is no finest-mesh floor: both named errors continue to quarter, so the fitted order 2.00 is not contaminated by round-off.

2.3 Boundary Treatment

Every neighbour that lands on $\partial\Omega$ is removed from the unknown list. If the stencil at interior index k has a Dirichlet neighbour with value g_{edge}, the term $a_{\text{face}}(U_k - g_{\text{edge}})/h^2$ contributes

$$b_k \mathrel{+}= \frac{a_{\text{face}}\, g_{\text{edge}}}{h^2} \tag{7}$$

to the right-hand side. For P5 ($u^* = e^x \sin(\pi y)$, $a = 1$) the west column $i = 0$ is the row set $k = 0, \ldots, n-1$; an example is

$$b[0] \mathrel{+}= \frac{e^0 \sin(\pi y_0)}{h^2}.$$

Replacing a Dirichlet row by e_k while leaving the corresponding column untouched would destroy symmetry [8]. Boundary nodes are therefore absent from A_h.

For the hand-checked small matrix ($n = 2$, $h = 1/3$, $a \equiv 1$, $k = 2i + j$), the four interior nodes form

$$A_h = \frac{1}{h^2}\begin{pmatrix} 4 & -1 & -1 & 0 \\ -1 & 4 & 0 & -1 \\ -1 & 0 & 4 & -1 \\ 0 & -1 & -1 & 4 \end{pmatrix} = \begin{pmatrix} 36 & -9 & -9 & 0 \\ -9 & 36 & 0 & -9 \\ -9 & 0 & 36 & -9 \\ 0 & -9 & -9 & 36 \end{pmatrix}, \qquad \operatorname{nnz}(A_h) = 12, \tag{8}$$

which matches `test_hand_matrix_n2` in `validation_tests.py`.

2.4 Discrete Structure, Stability, and Convergence

After Dirichlet elimination, every off-diagonal entry of A_h is non-positive and the row sums on interior stencils that do not touch the boundary are zero; boundary-adjacent rows are strictly diagonally dominant because a positive a_{face}/h^2 has been moved to b. With positive shared face coefficients, A_h is a symmetric M-matrix, hence SPD, and the discrete maximum principle holds: a discrete solution of $A_h U = 0$ with Dirichlet data in $[m, M]$ remains in $[m, M]$.

The continuous energy identity for (1) with homogeneous Dirichlet data is

$$\int_\Omega a|\nabla u|^2 \,\mathrm{d}x = \int_\Omega f u \,\mathrm{d}x, \tag{9}$$

so coercivity is controlled by $a_{\min} > 0$. On the grid, the shared-face assembly is the discrete analogue: $U^\top A_h U = \sum_{\text{faces}} a_{\text{face}}(\delta U)^2/h^2 \geq a_{\min}\,|U|^2_{H^1_h}$. Dirichlet elimination does not destroy that identity because boundary values enter b, not the quadratic form on interior unknowns. Consequently A_h remains SPD for every admissible a used in this report (smooth $a = 1+x$ and both harmonic/arithmetic layered averages).

Why the harmonic mean is forced at a jump. Across a vertical interface the exact flux $q = a u_x$ is continuous. On one grid edge of length h that straddles a_L and a_R in equal halves, integrating $u_x = q/a$ gives

$$U_R - U_L = q \cdot \tfrac{h}{2}\Big(\tfrac{1}{a_L} + \tfrac{1}{a_R}\Big) \quad \Rightarrow \quad q = \frac{2a_L a_R}{a_L + a_R}\,\frac{U_R - U_L}{h}. \tag{10}$$

So the face coefficient that preserves the continuous flux is exactly (6). The arithmetic mean $(a_L + a_R)/2$ is a different resistance and cannot match q unless $a_L = a_R$.

Convergence on smooth data. I derive the bound for P4 directly rather than quoting an equivalence theorem. Write $e = U - u^*|_{\text{grid}}$; since $A_h u^* = b - \tau_h$, the error solves $A_h e = \tau_h$, so any bound on A_h^{-1} converts the truncation of Table 3 into an error bound.

Lemma 1 (Barrier bound)**.** *Let A be a nonsingular M-matrix, so $A^{-1} \geq 0$ entrywise. If $\varphi \geq 0$ satisfies $A\varphi \geq c\mathbf{1}$ with $c > 0$, then $\|A^{-1}\|_\infty \leq \|\varphi\|_\infty / c$.*

Proof. Because $A^{-1} \geq 0$, $\|A^{-1}\|_\infty = \max_k (A^{-1}\mathbf{1})_k$ and $A^{-1}\mathbf{1} \leq c^{-1}A^{-1}(A\varphi) = \varphi/c$ componentwise. □

Theorem 1 (Second-order convergence for P4)**.** *For $a = 1 + x$ with half-point face coefficients, $\|A_h^{-1}\|_\infty \leq 1/4$ and $\lambda_{\min}(A_h) \geq 8\sin^2(\pi h/2)/h^2$ (≥ 19.68 for $n \geq 15$). Hence*

$$\|e\|_\infty \leq \tfrac{1}{4}\|\tau_h\|_\infty = \mathcal{O}(h^2), \qquad h\|e\|_2 \leq \frac{h\|\tau_h\|_2}{\lambda_{\min}(A_h)} = \mathcal{O}(h^2).$$

Proof. Take $\varphi(x) = x(1-x)/2$ at the interior nodes, so $\|\varphi\|_\infty \leq 1/8$. Because φ is quadratic, $(\varphi_{i+1} - \varphi_i)/h = \varphi'(x_{i+1/2})$ exactly, so each x-face flux equals $q(x_{i+1/2})$ with $q = (1+x)(\frac{1}{2} - x)$; q is quadratic, so its centred difference is exact and the x-part of row (i,j) equals $-q'(x_i) = \frac{1}{2} + 2x_i$. The y-differences of φ vanish. Where a neighbour is a Dirichlet node, elimination drops the term $-a_{\text{face}}\varphi_{\text{nbr}}/h^2 \leq 0$ from the row (on $x = 0, 1$ it is zero since $\varphi = 0$ there), which can only increase $(A_h\varphi)_{ij}$. Hence $A_h\varphi \geq \frac{1}{2}\mathbf{1}$, and Lemma 1 gives $\|A_h^{-1}\|_\infty \leq (1/8)/(1/2) = 1/4$. For L^2, the quadratic form $U^\top A_h U = \sum_{\text{faces}} a_{\text{face}}(\delta U)^2/h^2$ with $a_{\text{face}} \geq 1$ dominates the same sum for $-\Delta_h$, whose smallest eigenvalue is $8\sin^2(\pi h/2)/h^2$; then $\lambda_{\min}\|e\|_2^2 \leq e^\top A_h e = e^\top \tau_h \leq \|e\|_2\|\tau_h\|_2$. Both constants are h-independent and $\|\tau_h\| = \mathcal{O}(h^2)$ by the consistency argument above. □

Table 1 compares the predicted bounds with the observed errors. The max-norm bound is about $8\times$ pessimistic because τ_h peaks near $\partial\Omega$, where e is pinned to zero. With the computed $\lambda_{\min} \approx 28.7$ the L^2 bound is within 10% of the observed error, because $u^* = \sin(\pi x)\sin(\pi y)$ is close to the lowest eigenmode. All bounds shrink by 4 per refinement, matching the fitted order 2.00 of Table 2. `test_p4_barrier_bound` checks $A_h\varphi \geq 1/2$ and $\max(A_h^{-1}\mathbf{1}) \leq 1/4$ numerically.

Table 1: P4: proven bounds of Theorem 1 against observed errors. "rigorous" uses $\lambda_{\min}(-\Delta_h)$; "computed" uses $\lambda_{\min}(A_h)$ from `eigsh`.

n	$\frac{1}{4}\|\tau_h\|_\infty$	observed L^∞	L^2 bound (rigorous)	L^2 bound (computed)	observed L^2
15	2.57×10^{-2}	3.21×10^{-3}	2.53×10^{-3}	1.74×10^{-3}	1.60×10^{-3}
31	6.43×10^{-3}	8.01×10^{-4}	6.33×10^{-4}	4.35×10^{-4}	3.99×10^{-4}
63	1.61×10^{-3}	2.00×10^{-4}	1.58×10^{-4}	1.09×10^{-4}	9.97×10^{-5}
127	4.02×10^{-4}	5.01×10^{-5}	3.96×10^{-5}	—	2.49×10^{-5}

Exactness of the harmonic rule. On the layered jump the regularity assumptions of Theorem 1 fail, so I analyse the two face rules separately.

Proposition 1. *For the layered problem with odd N and harmonic interface coefficient* (6)*, the discrete solution equals u^* at every node.*

Proof. Insert u^* from (3) into (5), with the Dirichlet values $g = u^*$ treated as neighbours. u^* is independent of y, so all y-face fluxes vanish. An x-face that does not cross $x = 1/2$ has constant a and linear u^*, so its discrete flux equals $au_x^* = 20/11$ exactly; the interface face gives $20/11$ by (10). Every row is therefore a difference of equal fluxes, $(A_h u^*)_{ij} = 0 = f$, so $\tau_h \equiv 0$, and since A_h is SPD, $U = u^*$. □

First-order error of the arithmetic rule. Replace the interface coefficient by $a_A = (1+10)/2 = 11/2$ and look for a y-independent discrete solution. Since $f = 0$, every x-face carries the same discrete flux q_h. With interface columns $x_L = \frac{1}{2} - \frac{h}{2}$, $x_R = \frac{1}{2} + \frac{h}{2}$, the left slope is q_h, the right slope is $q_h/10$, and the interface jump is $U_R - U_L = q_h h/a_A$. Imposing $U(0) = 0$, $U(1) = 1$:

$$q_h\Big[\tfrac{11}{20}(1-h) + \tfrac{h}{a_A}\Big] = 1 \implies q_h = \frac{20/11}{1 - \frac{81}{121}h} = \frac{20}{11}\Big(1 + 0.669\,h + \mathcal{O}(h^2)\Big). \tag{11}$$

Setting $a_A = a_H = 20/11$ in (11) returns $q_h = 20/11$ exactly, which is Proposition 1 again. The nodal error is largest at x_L: $\|e\|_\infty = (q_h - \frac{20}{11})(\frac{1}{2} - \frac{h}{2}) \approx 0.61\,h$, i.e. 1.942×10^{-2}, 9.609×10^{-3}, 4.779×10^{-3}, 2.383×10^{-3} for $N = 31, 63, 127, 255$. The 2D values in Table 5 agree to 0.19%, 0.09%, 0.05%, 0.02%; the residual gap comes from the top and bottom Dirichlet rows, which carry the exact u^* rather than this discrete profile (`test_arithmetic_closed_form`). The same argument predicts a mid-line flux error $q_h - 20/11 = \mathcal{O}(h)$. Near $y = 0, 1$ the exact Dirichlet data are incompatible with the $\mathcal{O}(h)$-shifted interior profile across one cell, so the flux error there stays ≈ 0.66 under refinement.

Three numerical checks corroborate the derivation: $\|A - A^\top\|_{\max}/\|A\|_{\max} = 0$ on every assembled matrix in this report; the smallest algebraic eigenvalue of the P4 matrix at $n = 15$ is $\lambda_{\min} = 28.61 > 0$; the layered numerical solution stays in $[0, 1]$. Conditioning is separate: $\kappa_2(A_h)$ on P4 grows like h^{-2} (126.5, 529.7, 2.18×10^3 at $n = 15, 31, 63$), as expected for a second-order elliptic operator.

3 Implementation

3.1 Algorithm

Algorithm 1 Assembly of $A_h U = b$ and the direct solve

Require: interior count n, coefficient a, data f, g, face rule (half-point / harmonic / arithmetic)
Ensure: u_h, residual $\|b - A_h u_h\|_\infty$

```
1:  h ← 1/(n + 1); index k = in + j
2:  b ← f sampled at interior nodes
3:  for each interior node (i, j) do
4:      for each of the four faces do
5:          compute a shared a_face > 0
6:          if the neighbour is interior then
7:              add a_face/h^2 to the diagonal and −a_face/h^2 to that column
8:          else
9:              add a_face/h^2 to the diagonal and a_face g_edge/h^2 to b_k
10:         end if
11:     end for
12: end for
13: convert COO triplets to CSR; u_h ← spsolve(A, b)
14: report ‖b − A u_h‖_∞
```

Lines 4–11 build one shared face at a time so that A_h inherits symmetry from the continuous flux form; the Dirichlet branch (lines 9–10) is exactly (7).

3.2 Code Structure

File	Purpose
`model.py`	$a(x, y)$, manufactured u^*, layered exact profile (3)
`assemble.py`	sparse face-averaged stencil, index map, interface flux
`solvers.py`	`spsolve`, residual-stopped CG, AMG-preconditioned CG, $\lambda_{\min}$ / κ via `eigsh`
`verify.py`	named-norm errors, four-mesh tables, layered diagnostics
`plots.py`	every report figure
`fem1d.py` / `fem2d_oracle.py`	independent 1D P1 and optional 2D P1 oracles
`test_problem_bank.py`	analytic bank (P1–P5), self-check of $f = -\nabla \cdot (a \nabla u^*)$
`validation_tests.py`	sign, boundary, order, nodal-exact, and M-matrix checks
`run_all.py`	regenerates every figure and prints the quoted numbers

3.3 Key Design Decisions

I assemble in COO using vectorised 2D index masks over the four stencil directions (no Python loops over grid nodes) and convert to CSR once. The physics (a, f, g) lives in `model.py`, so the same assembler runs P4, P5 and the layered jump. The default solve is sparse direct: at matched accuracy it is the cheapest solver from $n = 127$ on and costs milliseconds below that (Table 6); unpreconditioned CG serves as a residual-stopped iterative check and a contrast-sensitivity probe, and AMG-CG as the scalable alternative. Because shared face coefficients make A_h symmetric positive definite, Krylov CG is preferred over stationary Jacobi or Gauss–Seidel (error contraction governed by $(\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$ rather than $(\kappa-1)/(\kappa+1)$). Cost measure (named before it is quoted): wall-clock solve time to a matched discrete L^2 target, with iteration counts as the secondary measure; I do not count assembly, eigenvalue estimates, or plotting.

4 Verification and Numerical Experiments

4.1 Verification: Method of Manufactured Solutions

The bank's finite-difference self-check on $f = -\nabla \cdot (a\nabla u^*)$ passes with worst relative error 5.8×10^{-6} (analytic data check, not a solver test) [8]. Sign and boundary gates come first: P2 ($u^* = x^2 - y^2$) has $L^\infty = 4.4 \times 10^{-16}$ at $n = 15$ (exact on this quadratic); P5 recovers the Tutorial 5 centre $u_h(0.5, 0.5) = 1.652$. Figure 1 shows those two fields in (x, y, u_h); both axes are drawn so the gate is not a 1D slice.

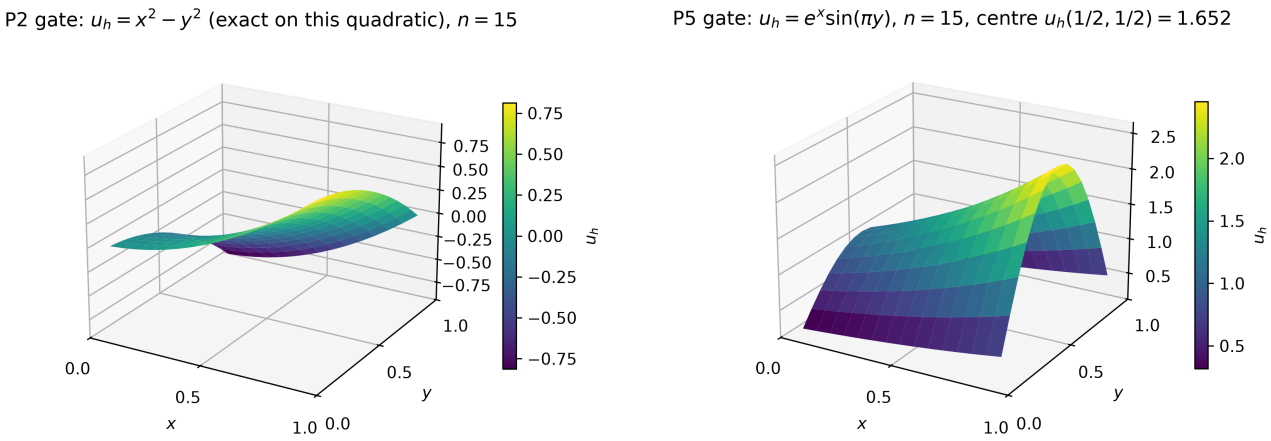
Table 2 is the P4 refinement study. Named norms: discrete L^2 given by $h\|u_h - u^*\|_2$, and $\|u_h - u^*\|_\infty$. Both fitted log-log slopes on $n = 15, 31, 63, 127$ are 2.00 (standard error of the fit $< 10^{-3}$). Figure 4 plots that table; Figure 2 is the P4 field as a surface; Figure 3 shows the sparsity pattern after elimination.

Table 2: P4 smooth manufactured solution, face coefficient at the half-point, `spsolve`. Discrete L^2 means $h\|u_h - u^*\|_2$.

n	h	L^2	L^∞	$\|b - Au_h\|_\infty$	p_{L^2}	p_∞
15	1/16	1.598×10^{-3}	3.205×10^{-3}	4.48×10^{-13}	—	—
31	1/32	3.988×10^{-4}	8.009×10^{-4}	2.90×10^{-12}	2.00	2.00
63	1/64	9.966×10^{-5}	2.003×10^{-4}	1.52×10^{-11}	2.00	2.00
127	1/128	2.491×10^{-5}	5.007×10^{-5}	6.84×10^{-11}	2.00	2.00

Table 3: P4 truncation $h\|b - A_h u^*\|_2$. Each refinement halves h and multiplies the truncation by $\approx 1/4$.

n	h	$h\|\tau_h\|_2$	$\|\tau_h\|_\infty$
15	1/16	4.978×10^{-2}	1.028×10^{-1}
31	1/32	1.248×10^{-2}	2.572×10^{-2}
63	1/64	3.122×10^{-3}	6.433×10^{-3}
127	1/128	7.808×10^{-4}	1.609×10^{-3}



(a) P2, $u_h = x^2 - y^2$, $n = 15$. **(b)** P5, $u_h = e^x \sin(\pi y)$, $n = 15$.

Figure 1: Bank gates as surfaces on (x, y). P2 is the quadratic sign check (nodally exact). P5 is the Tutorial 5 non-zero Dirichlet check; the centre is $u_h(1/2, 1/2) = 1.652$.

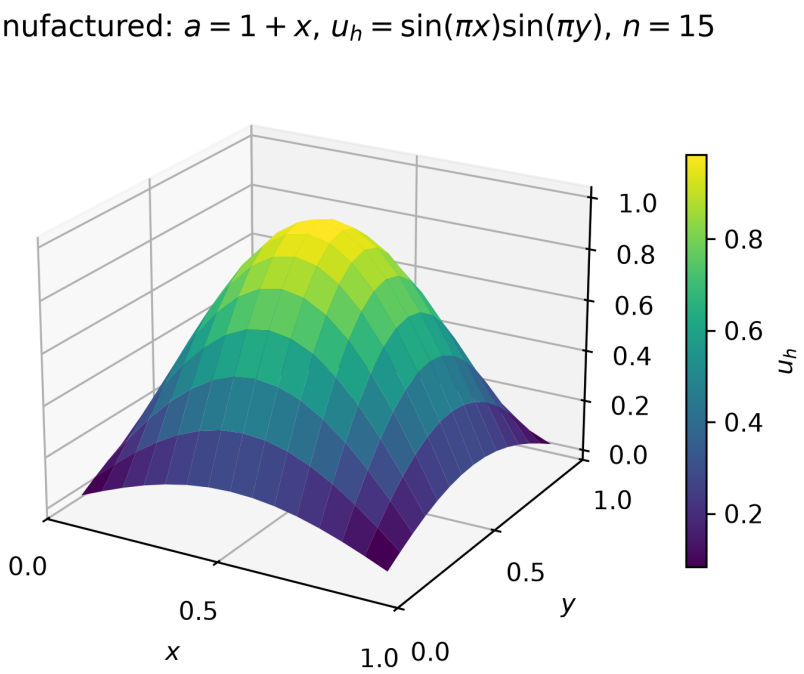


Figure 2: P4 manufactured field at $n = 15$: $a = 1 + x$, u_h approximates $\sin(\pi x)\sin(\pi y)$. Height is u_h; both x and y are labelled. Mesh convention: `indexing='ij'`, axis 0 is x.

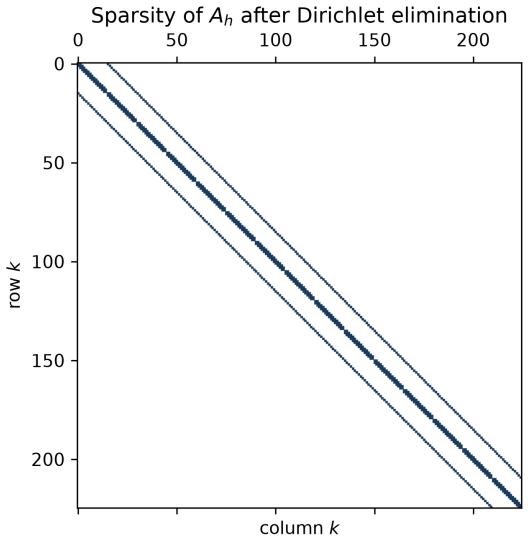


Figure 3: Spy plot of A_h at $n = 15$ after Dirichlet elimination (225×225). Five-point bandwidth matches the index map $k = in + j$.

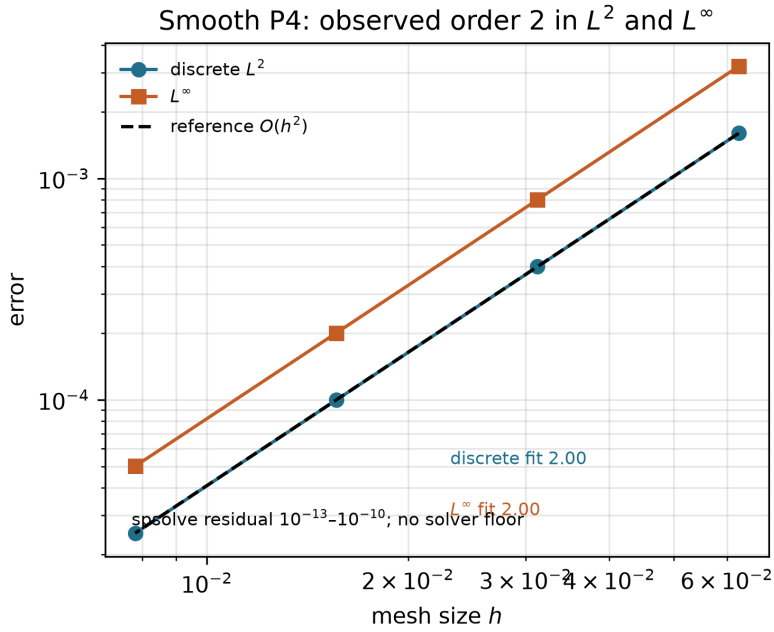


Figure 4: Log-log P4 errors versus h. Fitted discrete-L^2 order 2.001 ± 0.000, L^∞ order 2.000 ± 0.000, four meshes, reference slope 2. Confirms that the face-averaged stencil is second order on smooth data.

An independent 2D P1 FEM solve of the same P4 data (`scikit-fem`, homogeneous Dirichlet by `condense`) gives $L^2 = 5.18 \times 10^{-3}$ at $h_{\max} = 8.84 \times 10^{-2}$, residual 2×10^{-15}: an order-of-magnitude cross-check on a different mesh and method. The 1D P1 training problem $-u'' = \pi^2 \sin(\pi x)$ with nodal trapezoidal load fits discrete-L^2 order 2.00; the hand-checked element matrix is

$$K^{(e)} = \frac{1}{h}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad h = \tfrac{1}{4} \Rightarrow \begin{pmatrix} 4 & -4 \\ -4 & 4 \end{pmatrix}.$$

4.2 Separating Algebraic and Discretisation Error

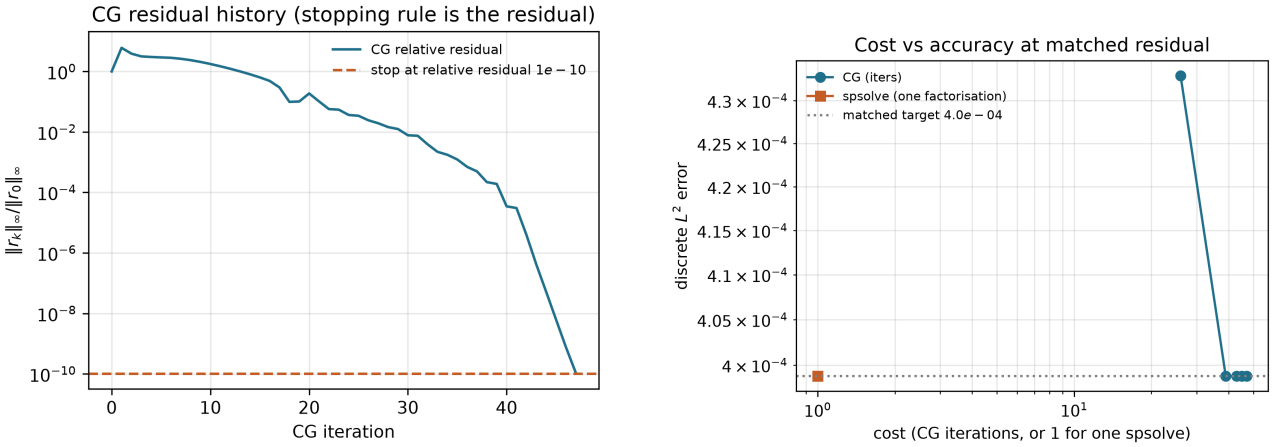
`spsolve` has no tolerance parameter. I treat its solution as the exact discrete solution and report the backward residual. On P4 at $n = 31$, $\|b - Au_h\|_\infty = 2.90 \times 10^{-12}$ while the discrete L^2 error is 3.99×10^{-4}. A crude forward-error sketch $\|e\|_2 \lesssim \|r\|_2/\lambda_{\min}$ with $\lambda_{\min} \approx 28.7$ puts the algebraic error

near 10^{-13}, four orders below the discretisation error.

Unpreconditioned CG on the same matrix, stopped on the relative residual $\|r_k\|_\infty/\|r_0\|_\infty$, is the second demonstration (Figure 5). At relative tolerance 10^{-2} the discrete L^2 error is still 4.33×10^{-4}; from 10^{-4} onwards it has locked onto the direct value 3.988×10^{-4} (Table 4). Tightening further does not change the reported discretisation error, so the fitted order in Table 2 is the discretisation rate. I stop CG on the residual, never on $\|x^{k+1} - x^k\|$.

Table 4: P4, $n = 31$: CG versus `spsolve`. Cost for CG is iteration count; `spsolve` is one factorisation. The L^2 column is the quantity of interest; wall times across meshes are in Table 6.

Solve	iterations	$\|b - Au\|_\infty$	discrete L^2
CG, rel. 10^{-2}	26	7.5×10^{-1}	4.329×10^{-4}
CG, rel. 10^{-4}	39	5.9×10^{-3}	3.988×10^{-4}
CG, rel. 10^{-6}	43	1.3×10^{-5}	3.988×10^{-4}
CG, rel. 10^{-10}	47	3.4×10^{-9}	3.988×10^{-4}
`spsolve`	—	2.9×10^{-12}	3.988×10^{-4}



(a) CG relative residual vs. iteration.

(b) Discrete L^2 vs. CG iterations.

Figure 5: Left: residual history with stopping line at 10^{-10}. Right: discrete L^2 versus CG iterations; dashed line is the `spsolve` error. Once the residual is tight enough the error floor is discretisation.

4.3 Depth Comparison: Harmonic versus Arithmetic Face Coefficients

This is the project's depth contribution beyond a smooth MMS: the same scheme, same grids, same Dirichlet data (3), same direct solver; only the face rule changes. The hypothesis under test is that "any positive face average that keeps A_h symmetric is an acceptable interface scheme." Table 5 and Figures 6–8 falsify that hypothesis and confirm (10).

Harmonic averaging is nodally exact: $L^\infty \le 3 \times 10^{-14}$ on all four meshes, and the numerical flux $a_E(U_R - U_L)/h$ equals 20/11 to at most 5×10^{-13} (the pack's sanity check) [8]. An interface-fitted 1D P1 FEM with 32 elements (so $x = 1/2$ is a node) reproduces the same profile to $L^\infty = 6.7 \times 10^{-16}$. Two methods that share neither the stencil nor the mesh convention agree on the exact nodal values.

Arithmetic averaging still produces a symmetric M-matrix, but the wrong series resistance in (10). The maximum flux jump stays ≈ 0.66 under refinement ($\mathcal{O}(1)$), but it sits in at most 8 rows next to the interface–boundary corners; on the mid-line $y \approx 1/2$ the flux error is 4.70×10^{-2}, 2.29×10^{-2}, 1.13×10^{-2}, 5.64×10^{-3}, first order as (11) predicts. The L^∞ solution error is first-order ($\approx 1.94 \times 10^{-2}$ down to 2.38×10^{-3} as N runs through $31, 63, 127, 255$) and matches the closed form below (11). Relative to the smooth-case energy (9), the flux error is the physically meaningful defect: arithmetic keeps a discrete energy form but with the wrong a_{face}, so L^2-looking plots can look stable while transport is wrong.

On every harmonic and arithmetic run the discrete solution stays in $[0, 1]$: both averages inherit the M-matrix / DMP. Monotonicity alone does not select the correct face rule. Value of the comparison: it separates algebraic structure (symmetry, SPD) from interface modelling (flux continuity), which is the core numerical lesson of Direction 3.

Table 5: Layered $a = 1\,|\,10$, odd interval counts so $x = 1/2$ bisects a grid edge. Flux jump is $\max_j |a_E(U_R - U_L)/h - 20/11|$; p_∞ is the observed arithmetic L^∞ order.

n	N	harm. L^∞	harm. flux jump	arith. L^∞	arith. p_∞	arith. flux jump
30	31	2.4×10^{-15}	3.9×10^{-14}	1.94×10^{-2}	—	0.663
62	63	7.9×10^{-15}	1.1×10^{-13}	9.60×10^{-3}	1.01	0.662
126	127	2.6×10^{-14}	1.8×10^{-13}	4.78×10^{-3}	1.01	0.662
254	255	3.5×10^{-14}	4.6×10^{-13}	2.38×10^{-3}	1.00	0.662

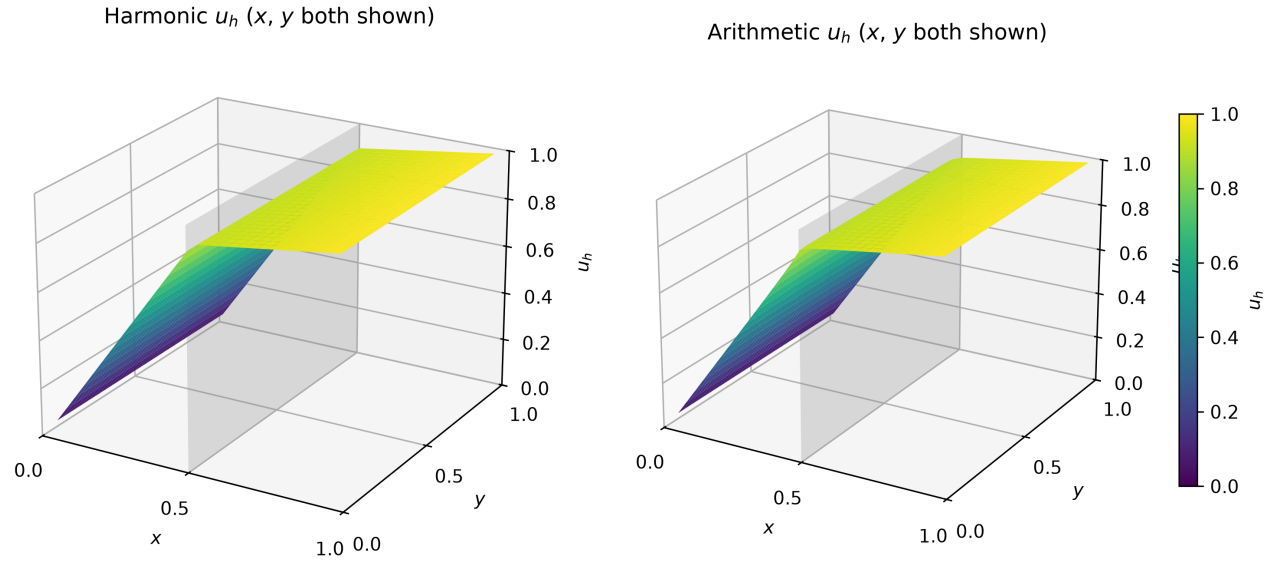
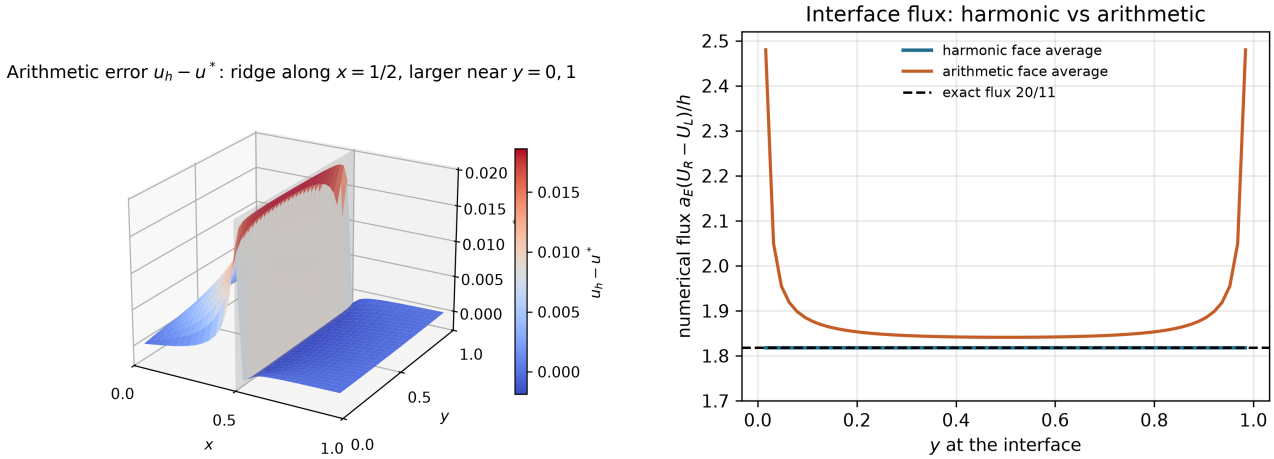


Figure 6: Layered u_h at $n = 30$ as surfaces over (x, y). Left: harmonic mean; the kink is the plane $x = 1/2$ and the field is independent of y. Right: arithmetic mean, same colour scale $[0, 1]$. The two face rules are compared on the full square, not on a single mid-line.



(a) Arithmetic error $u_h - u^*$.

(b) Interface flux versus y.

Figure 7: Where the arithmetic rule fails in (x, y). Left: error ridge along the interface $x = 1/2$, larger near $y = 0$ and $y = 1$. Right: numerical flux $a_E(U_R - U_L)/h$ along that same vertical face; harmonic sits on $20/11$, arithmetic is offset by $\mathcal{O}(h)$ in the interior and by ≈ 0.66 in the rows next to $y = 0, 1$.

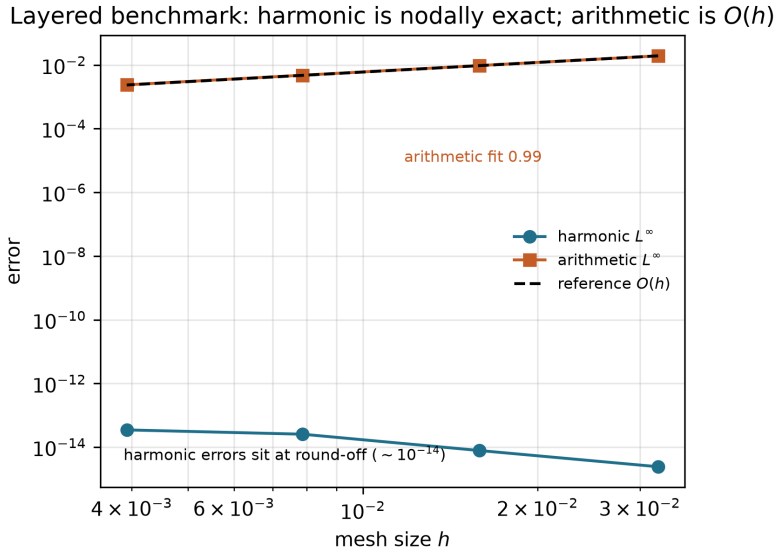


Figure 8: Layered L^∞ versus h. Harmonic errors sit at round-off ($\sim 10^{-14}$). Arithmetic tracks the $O(h)$ reference line (fitted slope 0.99). Same data, two face rules, named norm: the depth comparison.

4.4 Main Results: Contrast, Conditioning, Cost

Unpreconditioned CG iteration counts on the layered harmonic matrix at $n = 30$, relative residual 10^{-10}, grow with the contrast a_R/a_L: 125, 227, 402, 615 iterations at contrasts $2, 10, 50, 200$ (Figure 10). Coefficient contrast worsens conditioning but preserves the discrete maximum principle. On smooth P4, $\kappa_2(A_h)$ tracks h^{-2} as already quoted.

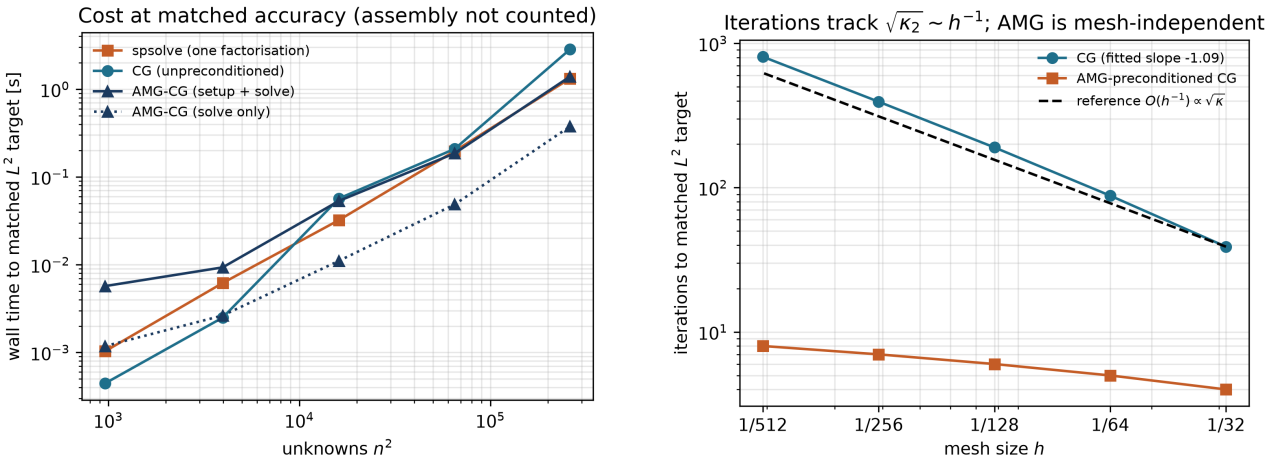
Cost–accuracy at a matched target. Cost measure (stated before quoting): wall-clock time of the solve phase, median of 5 repeats on one laptop, with CG and AMG-CG iteration counts as the hardware-independent secondary measure. For AMG-preconditioned CG (`pyamg` smoothed aggregation, Tutorial 6) the hierarchy setup is counted and reported separately from the solve. Not counted: assembly, eigenvalue estimates, figure rendering, and Python interpreter startup. Accuracy target (frozen across the comparison): on each P4 mesh, discrete L^2 error within 0.1% of the `spsolve` value on that mesh; each iterative solver gets the loosest relative residual in $\{10^{-4}, \ldots, 10^{-10}\}$ that meets it. At $n = 31$ this is Table 4: CG meets the target at relative residual 10^{-4} after 39 iterations. Table 6 and Figure 9 extend the comparison to $n = 511$ (2.6×10^5 unknowns).

Table 6: P4 cost at matched accuracy (discrete L^2 within 0.1% of `spsolve`). Wall times in seconds, median of 5 runs; assembly not counted. AMG-CG time is setup + solve.

n	unknowns	target L^2	`spsolve`	CG it.	CG time	AMG-CG it.	AMG-CG time
31	961	3.99×10^{-4}	0.0010	39	0.0004	4	$0.0045 + 0.0012$
63	3 969	9.97×10^{-5}	0.0062	88	0.0025	5	$0.0067 + 0.0026$
127	16 129	2.49×10^{-5}	0.032	190	0.057	6	$0.042 + 0.011$
255	65 025	6.23×10^{-6}	0.197	394	0.209	7	$0.137 + 0.049$
511	261 121	1.56×10^{-6}	1.32	807	2.85	8	$1.02 + 0.38$

Unpreconditioned CG is cheapest on the two coarsest meshes, where every solver finishes in milliseconds. Its iteration count roughly doubles per refinement (fitted slope -1.09 against h, Figure 9 right), as the CG bound $\sqrt{\kappa_2} \propto h^{-1}$ predicts from the $\kappa_2 \sim h^{-2}$ growth above; from $n = 127$ `spsolve` is cheaper, and at $n = 511$ CG is 2.2× slower. AMG-CG needs 4–8 iterations on every mesh and its solve phase alone is 3.5× faster than `spsolve` at $n = 511$, but the setup costs 1.02 s; setup plus solve is within 6% of `spsolve` at $n = 255$ and $n = 511$, comparable to the run-to-run timing scatter.

Conclusion (negative for both iterative solvers on this range): at matched accuracy no iterative solver beats one sparse factorisation reliably for $n \geq 127$ on this 2D problem. AMG-CG would pay off only if its hierarchy were reused for several right-hand sides or the mesh grew beyond $n = 511$, which is outside the meshes reported here.



(a) Wall time at matched L^2 versus unknowns. **(b)** Iterations to matched L^2 versus h.

Figure 9: P4 cost study, $n = 31$–511. Left: `spsolve`, unpreconditioned CG, and AMG-CG (solid: setup + solve; dotted: solve only); assembly not counted. Right: CG iterations follow the h^{-1} reference, while AMG-CG iterations are nearly mesh-independent.

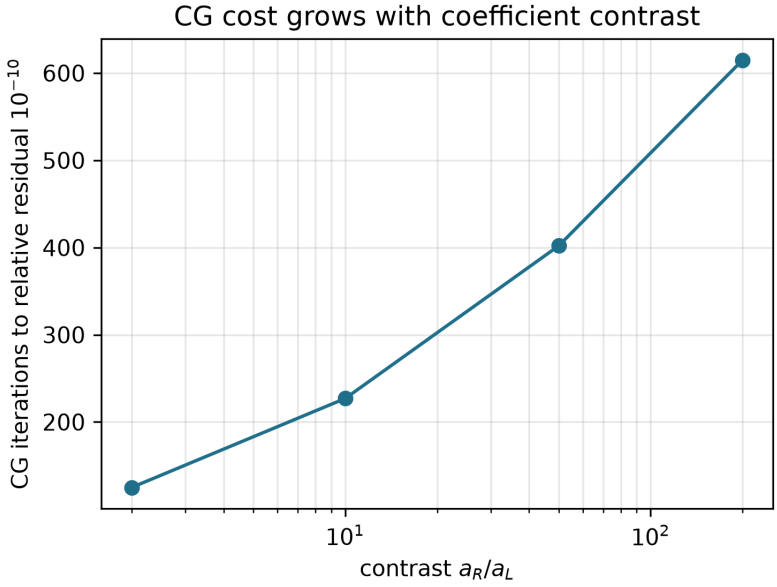


Figure 10: CG iterations versus conductivity contrast on the layered problem, $n = 30$, residual stop 10^{-10}. Cost measure: iterations. Contrast raises iterative cost without breaking the discrete maximum principle.

4.5 Limitations and Robustness

The second-order claim applies to smooth a with the half-point face rule. Across a jump, harmonic averaging on an aligned grid is exact for this piecewise-linear u^*. Arithmetic averaging remains an M-matrix and still looks smooth on a colour plot, yet its flux is wrong. Wall times are from one machine and scatter by roughly 10% between runs, so the AMG-CG versus `spsolve` ranking at $n \geq 255$ is a tie rather than a result; iteration counts are the robust part of the cost study. The proofs of Theorem 1 and Proposition 1 use the specific P4 coefficient and the aligned, y-independent layered solution; a non-aligned or curved interface is not covered.

5 Conclusions

- A conservative face-averaged five-point scheme for $-\nabla \cdot (a\nabla u)$ with Dirichlet data moved into b was assembled and verified on the analytic bank before the jump.
- On smooth P4, discrete L^2 and L^∞ both fit order 2.00 over four meshes; $\|A - A^\top\|_{\max} = 0$ and $\lambda_{\min} > 0$. A barrier function proves $\|e\|_\infty \le \|\tau_h\|_\infty/4$, and the eigenvalue bound predicts the L^2 error within 10%. The reported rate survives a CG tolerance sweep.
- On the layered jump, harmonic face coefficients are provably nodally exact and keep the flux at 20/11; for arithmetic coefficients the closed form (11) predicts the first-order L^∞ error to 0.2%, and the flux error is $\mathcal{O}(h)$ on the mid-line but $\mathcal{O}(1)$ at the corners. Symmetry is necessary but not sufficient for correct interface physics.
- Cost is wall time at matched L^2, with iterations secondary. CG wins only for $n \le 63$; `spsolve` wins from $n = 127$; AMG-CG has mesh-independent iterations but ties `spsolve` at $n = 511$ once setup is counted. Next step: non-aligned interfaces, and AMG reuse across several right-hand sides.

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[6] C. R. Harris et al., "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.

[7] T. Gustafsson and G. D. McBain, "scikit-fem: a Python package for finite element assembly," *Journal of Open Source Software*, vol. 5, no. 52, 2020.

[8] Course problem pack, "Variable diffusion," Project 2 Direction 3, Numerical Analysis of ODEs and PDEs, 2026.

[9] Course brief, "Numerical Solution of Elliptic Boundary Value Problems," Project 2, 2026.

A AI Transparency Log

I used Grok (via Cursor) to help write the Python code and to polish the English of this report.

Item	Claim / assumption	Risk / falsifier	Independent check & decision
1	Conservative stencil (5) with Dirichlet shift $b_k \mathrel{+}= a_{\text{face}} g/h^2$	Sign error in boundary shift or unsymmetric boundary row	Hand matrix (8) (nnz = 12), P2 $L^\infty = 4.4 \times 10^{-16}$, P5 centre 1.652; accepted
2	Layered exact profile (3) and harmonic vs. arithmetic face average	Using a discontinuous OCR formula or treating symmetry alone as proof of flux accuracy	Continuity of u and $au_x = 20/11$ checked analytically; Table 5 shows harmonic $L^\infty \sim 10^{-14}$ vs. arithmetic $O(h)$

Ownership and library interfaces: Self-implemented in `code/assemble.py`, `model.py`, `verify.py`, `fem1d.py`. Library calls: `scipy.sparse.coo_matrix` / `spsolve` (direct CSR solve), `scipy.sparse.linalg.cg` (residual-stopped CG), `eigsh` ($\lambda_{\min}, \kappa_2$), `pyamg.smoothed_aggregation_solver` (AMG preconditioner), and `scikit-fem` (2D P1 cross-check).

B Individual Contribution Statement

Individual Contribution Statement — Project 2 (PDE BVP)
Name: Chuyang Wang Student number: 2363591
Role this project: sole author (implementation, weekly tasks, report).
The specific things I did this project:

1. Derived the conservative stencil (5), the Dirichlet shift (7), and the $n = 2$ hand matrix (8); implemented vectorised COO assembly in `code/assemble.py` with the index map $k = in + j$ (cf. Appendix A, Item 1).

2. Completed the weekly individual challenges through Week 8: Week 5 five-point P5 centre $u_h(0.5, 0.5) = 1.652$; Week 6 four-mesh slope (2.00 in discrete L^2 and L^∞); Week 7 residual/tolerance check separating algebraic from discretisation error (Table 4); Week 8 feedback-to-evidence improvement on harmonic versus arithmetic interface flux (Table 5, Figures 6–8).

3. Re-derived the layered profile (3), implemented harmonic versus arithmetic face averages, and produced Table 5 and Figures 6–8 (cf. Appendix A, Item 2).

4. Proved the barrier bound (Theorem 1), harmonic exactness (Proposition 1) and the arithmetic closed form (11), and checked each against the code (Table 1, `validation_tests.py`).

5. Separated algebraic from discretisation error by a CG residual sweep (Table 4); measured CG iterations against contrast (Figure 10); ran the matched-accuracy cost study of `spsolve`, CG and AMG-CG up to $n = 511$ (Table 6).

6. Wrote `validation_tests.py` and `run_all.py`, and wrote the full LaTeX report including the AI log and this ICS.

What I would do differently: freeze the layered exact formula on paper before reading the OCR'd pack, instead of debugging a "failed" nodal-exact test against a discontinuous u^*.
Anything I should have asked for help with earlier: whether odd N (interface between columns) versus even N (interface on a node) is required for the FDM sanity check, before running the first layered table on $n = 15$.

C Supplementary data

P4 structure at $n = 15$: $\|A - A^\top\|_{\max}/\|A\|_{\max} = 0$, $\lambda_{\min} = 28.61$, $\kappa_2 \approx 1.27 \times 10^2$.
FEM 1D element matrix for $h = 1/4$: $\left(\begin{smallmatrix} 4 & -4 \\ -4 & 4 \end{smallmatrix}\right)$.
All numbers in the tables are reproduced by `python code/run_all.py`.